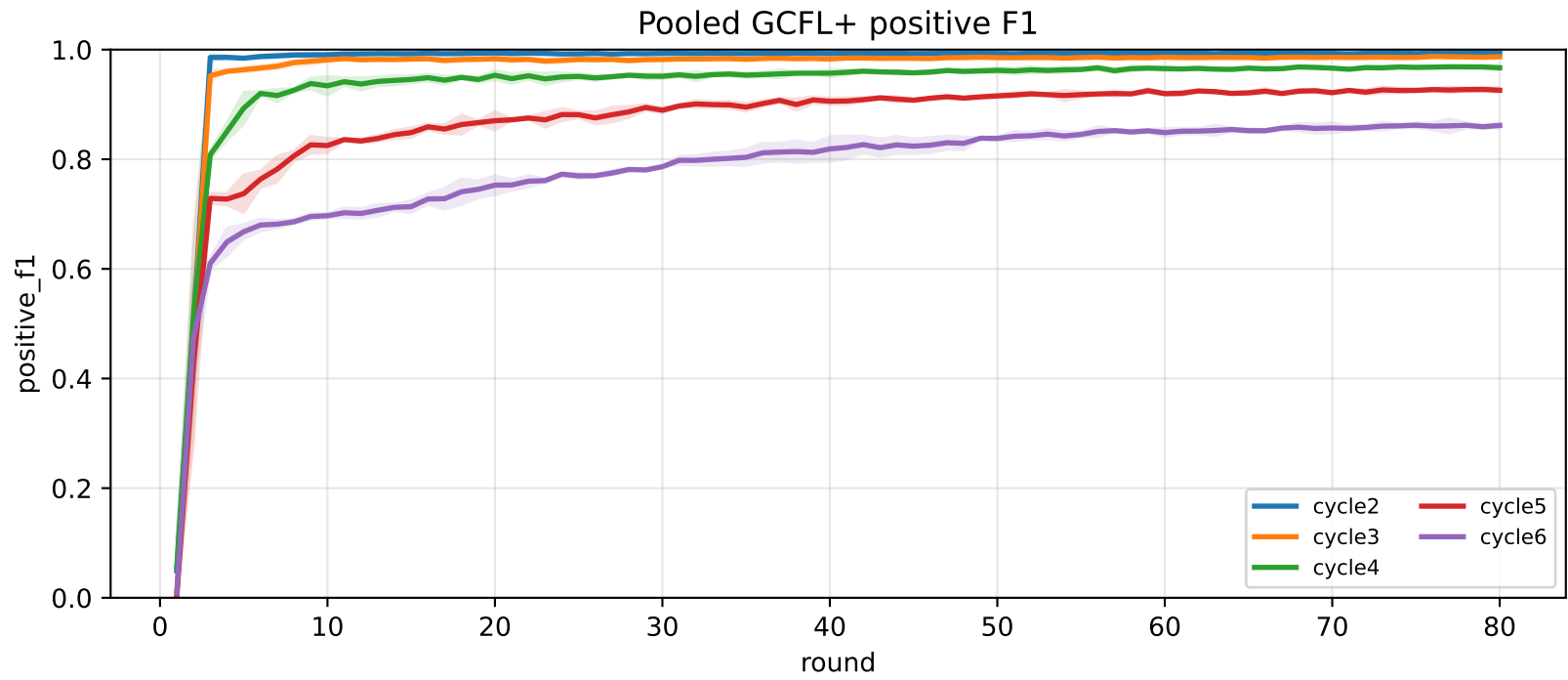
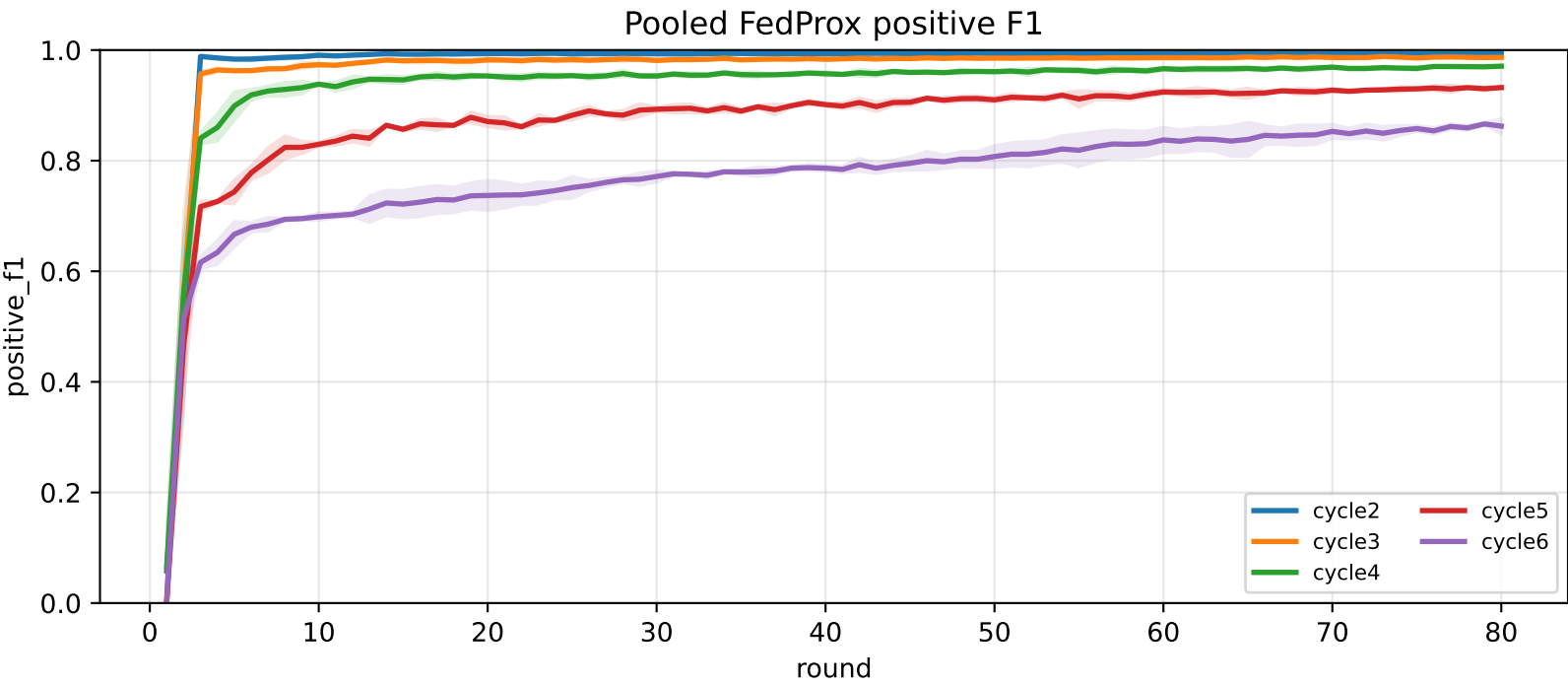
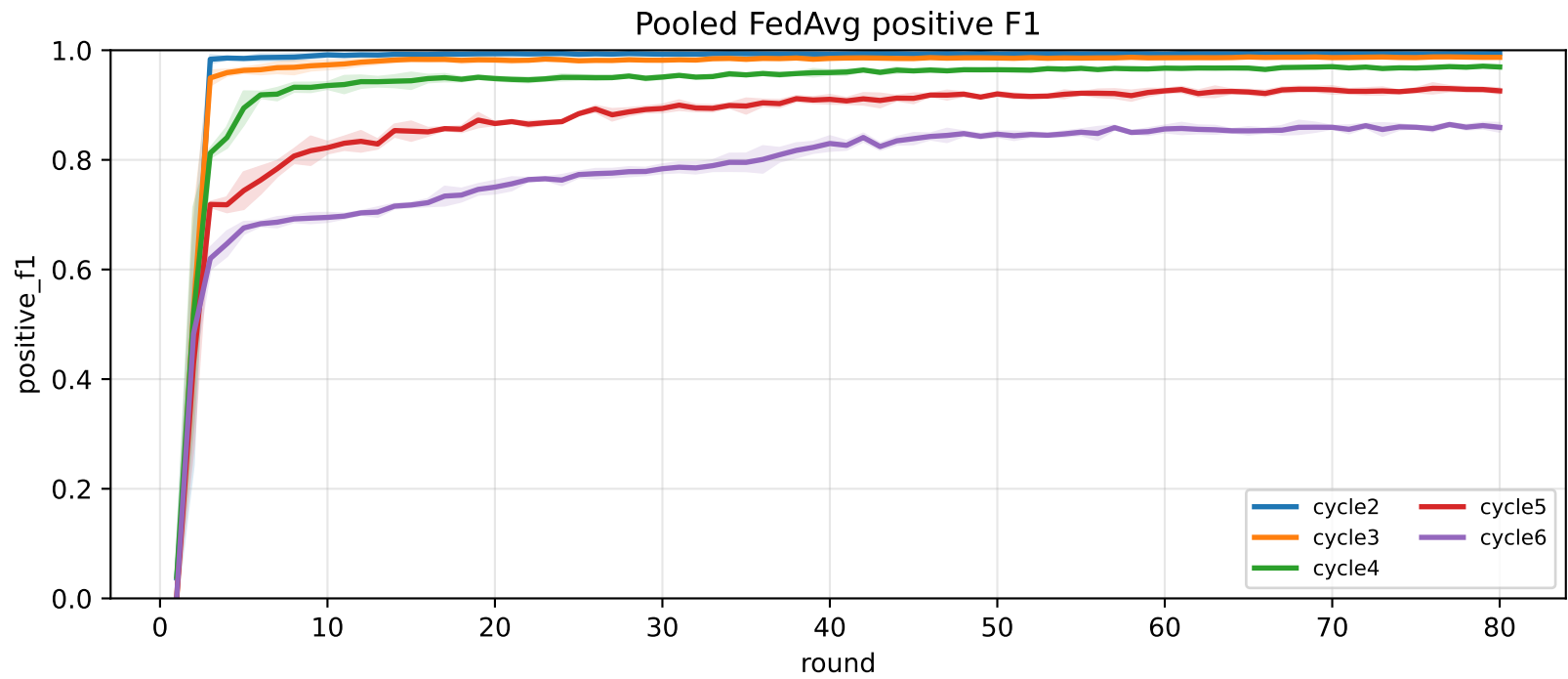
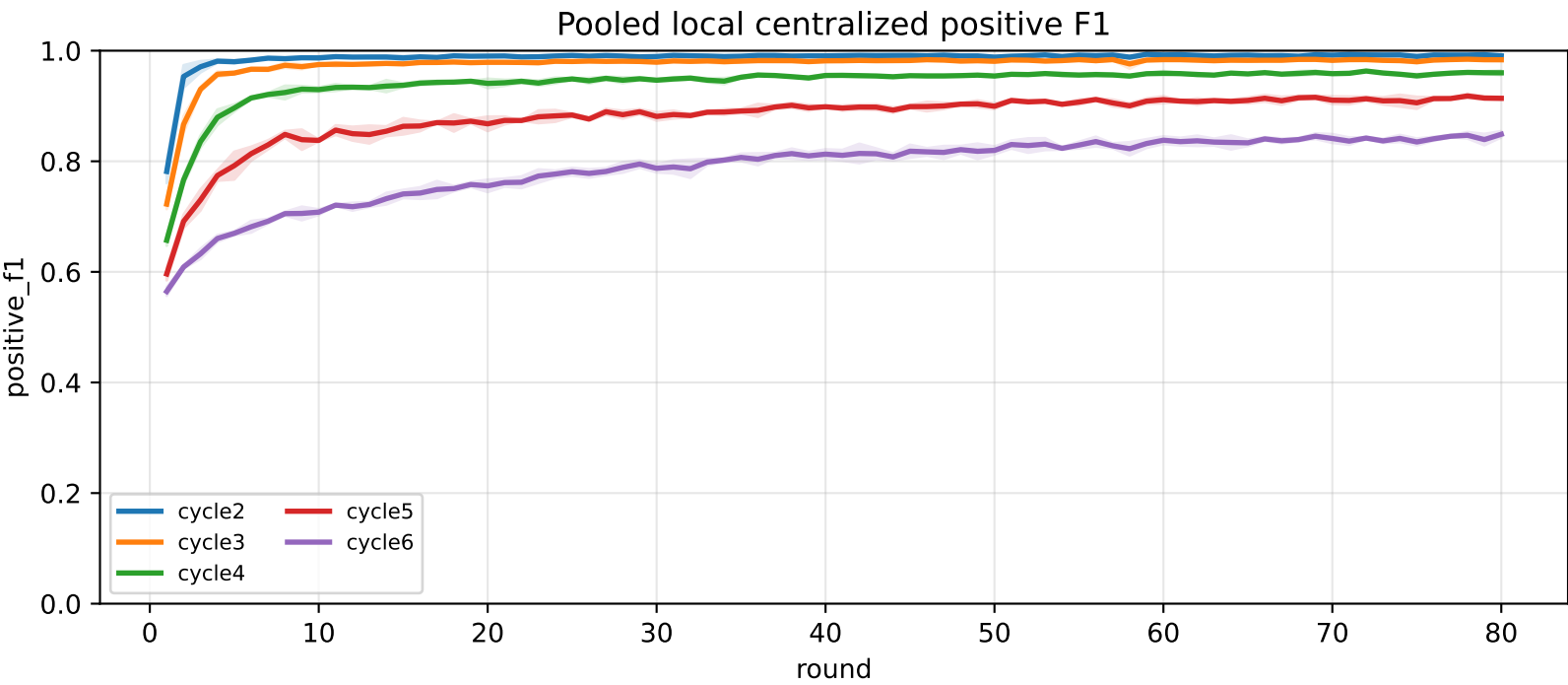
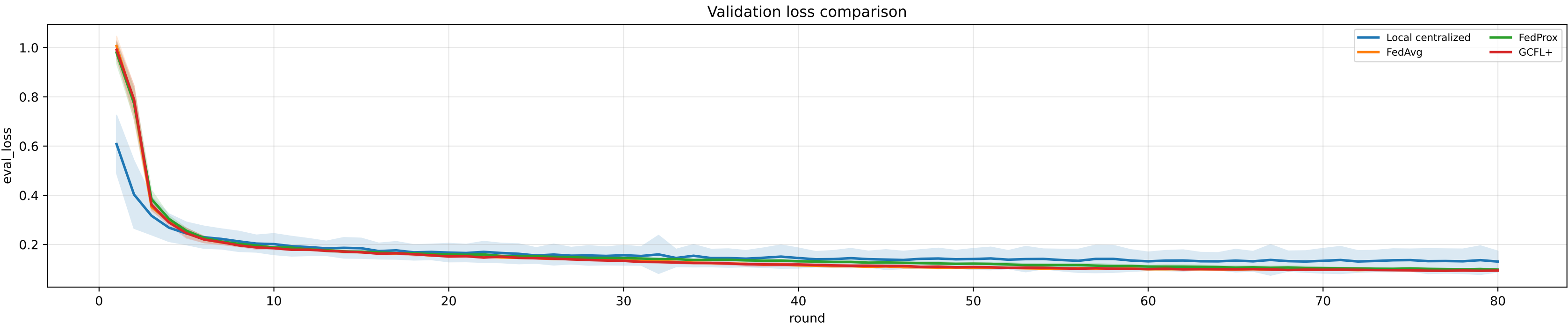


Heterogeneity level: High | aggregated / pooled performance over 20 clients |
model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64

heterogeneity=high | task_jsd_mean=0.028 | task_jsd_median=0.018 | task_jsd_max=0.120 | family_centroid_jsd_mean=0.033
per-task pos-rate std | cycle2_std=0.120 | cycle3_std=0.071 | cycle4_std=0.032 | cycle5_std=0.054 | cycle6_std=0.087

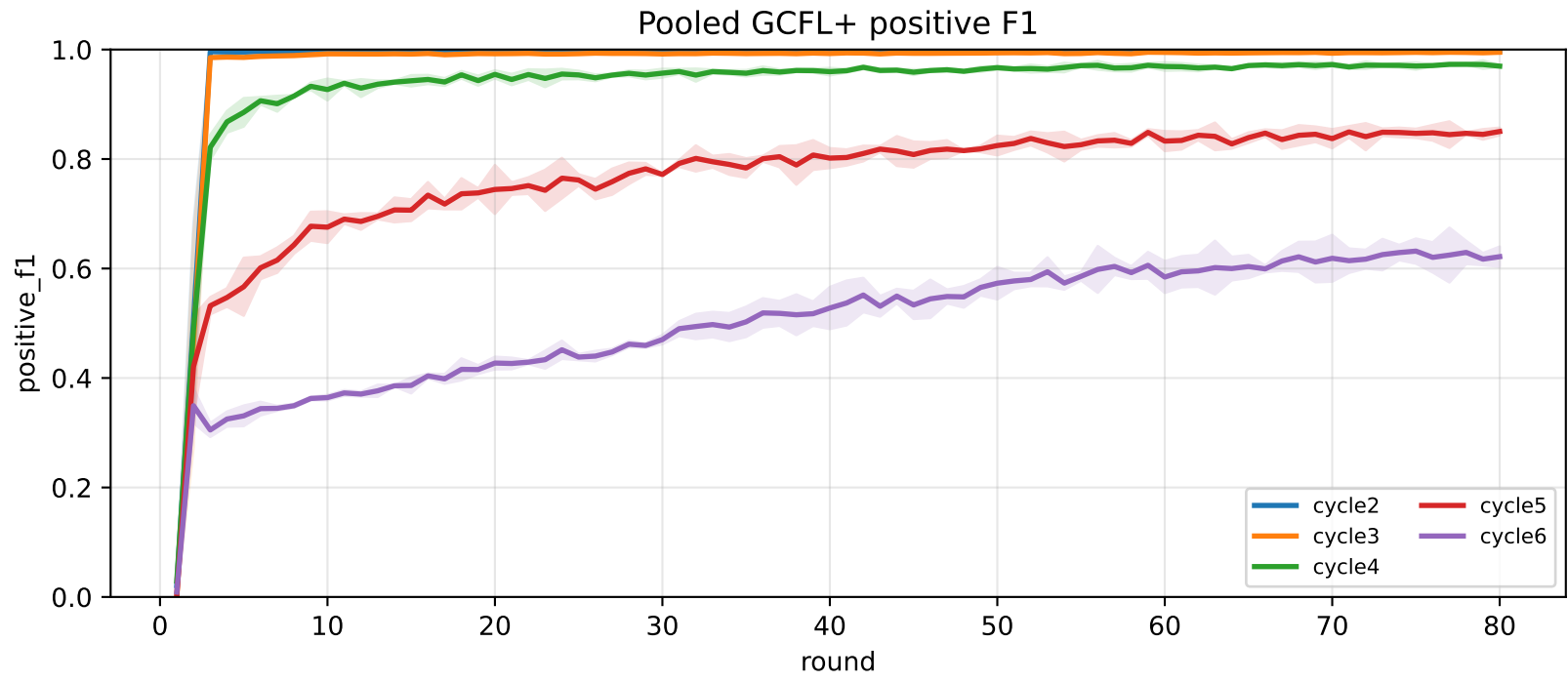
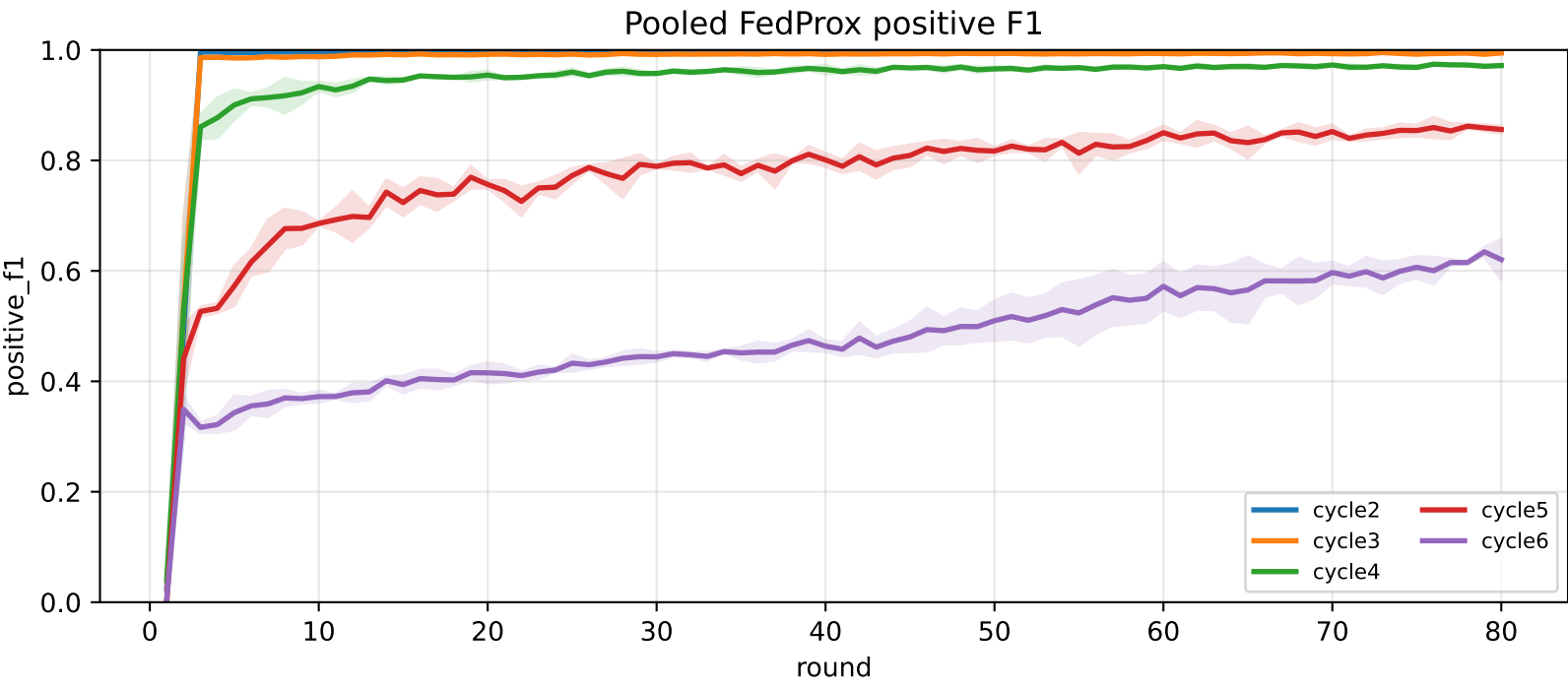
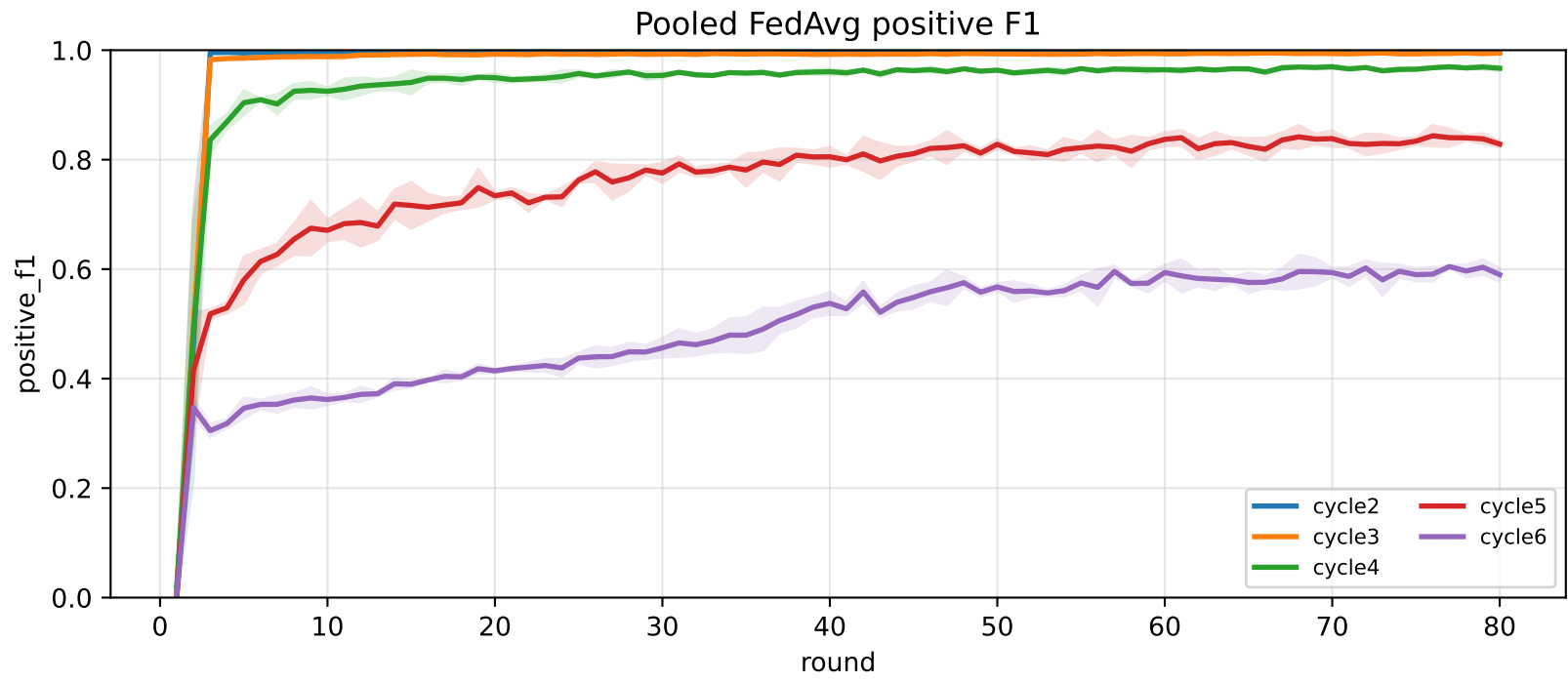
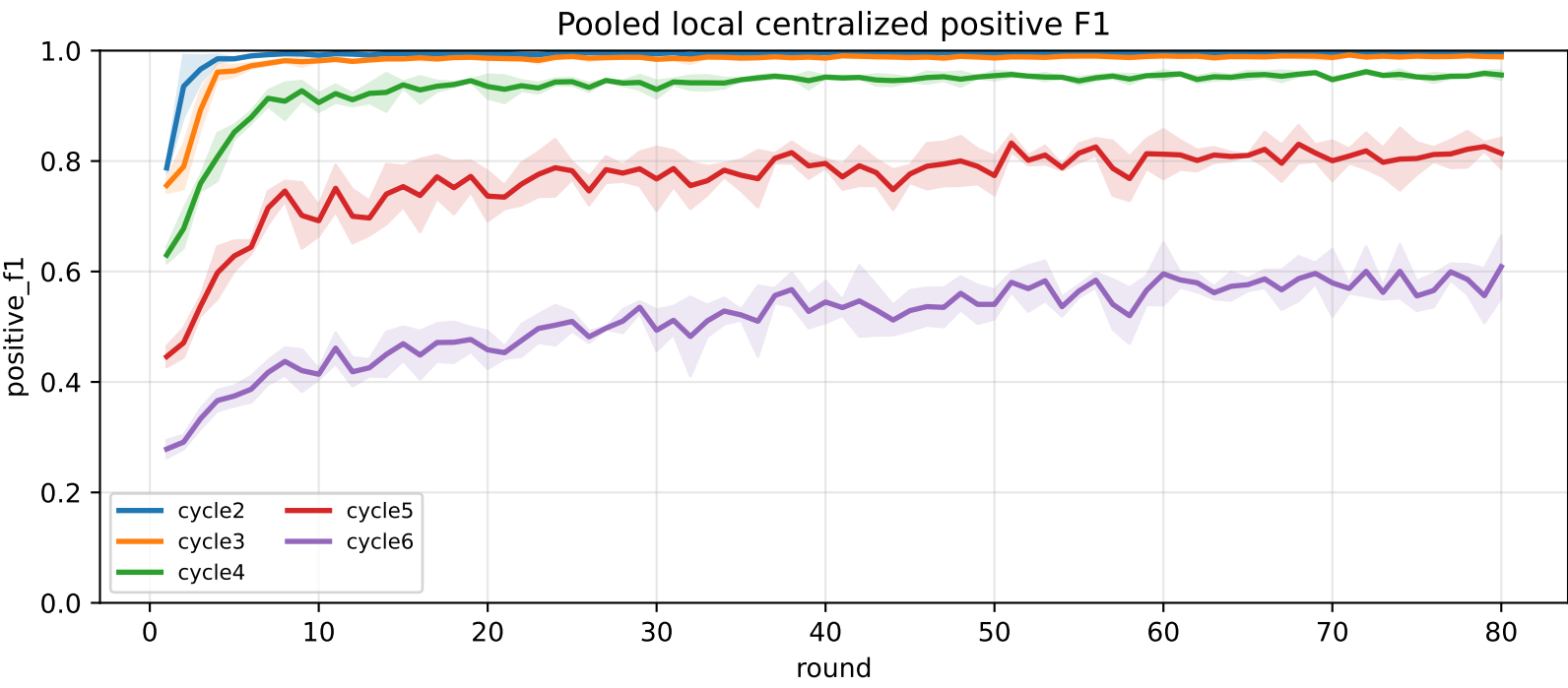
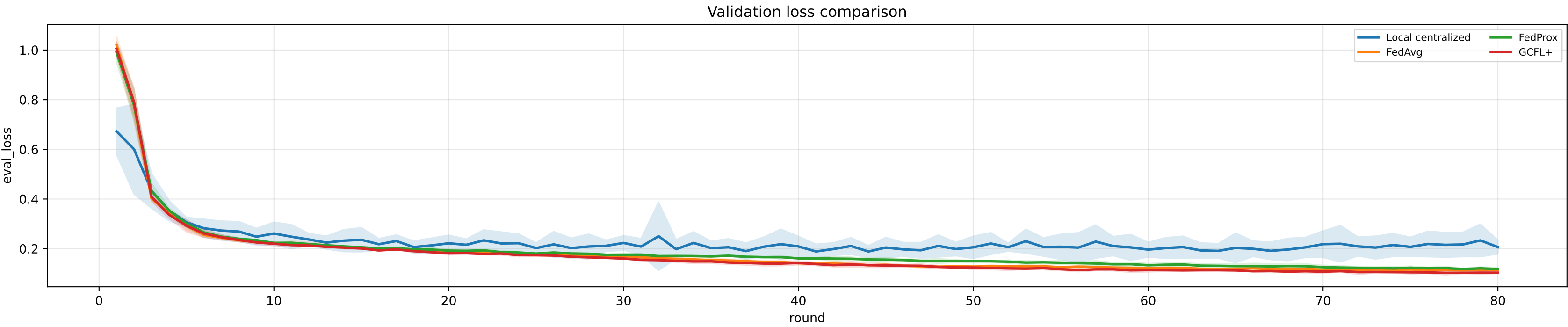
Best logged test performance (mean ± std over seeds)							
Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	93.8% ± 0.1%	92.8% ± 0.1%	99.3% ± 0.1%	98.5% ± 0.1%	96.1% ± 0.1%	91.4% ± 0.2%	84.2% ± 0.3%
FedAvg	95.0% ± 0.1%	94.1% ± 0.1%	99.4% ± 0.0%	98.7% ± 0.0%	96.9% ± 0.1%	93.5% ± 0.1%	86.7% ± 0.3%
FedProx	94.9% ± 0.3%	94.1% ± 0.3%	99.4% ± 0.0%	98.7% ± 0.0%	97.0% ± 0.2%	93.5% ± 0.3%	86.4% ± 1.2%
GCFL+	94.7% ± 0.1%	93.9% ± 0.1%	99.3% ± 0.0%	98.5% ± 0.0%	96.7% ± 0.1%	92.9% ± 0.0%	86.7% ± 0.3%



Heterogeneity level: High | family aggregated / pooled performance |
family=strong_decreasing | n_clients=4 |
model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64

family=strong_decreasing | cycle2=44.0%±0.3% | cycle3=34.3%±0.7% | cycle4=22.7%±0.7% | cycle5=13.9%±0.4% | cycle6=7.8%±0.6%
heterogeneity=high | task_jsd_mean=0.028 | task_jsd_median=0.018 | task_jsd_max=0.120 | family_centroid_jsd_mean=0.033
per-task pos-rate std | cycle2_std=0.120 | cycle3_std=0.071 | cycle4_std=0.032 | cycle5_std=0.054 | cycle6_std=0.087

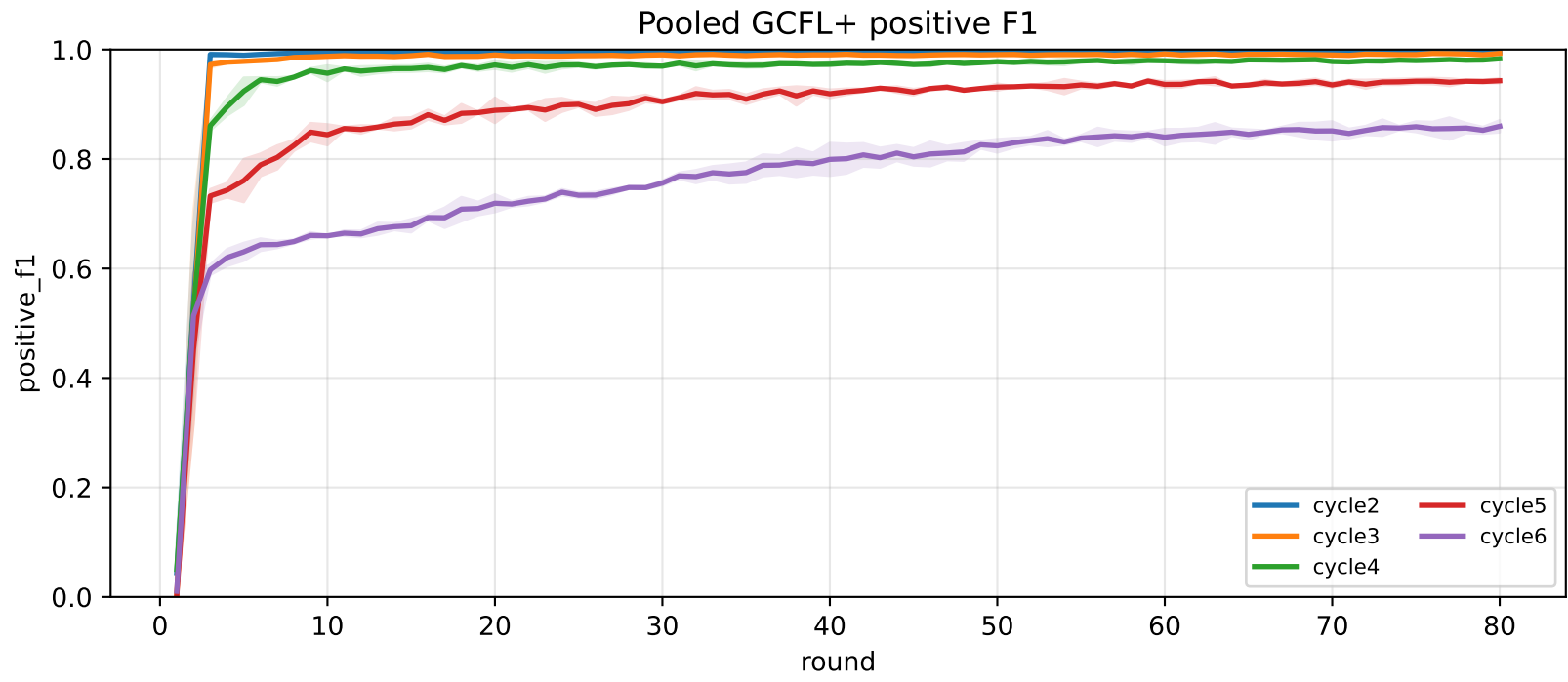
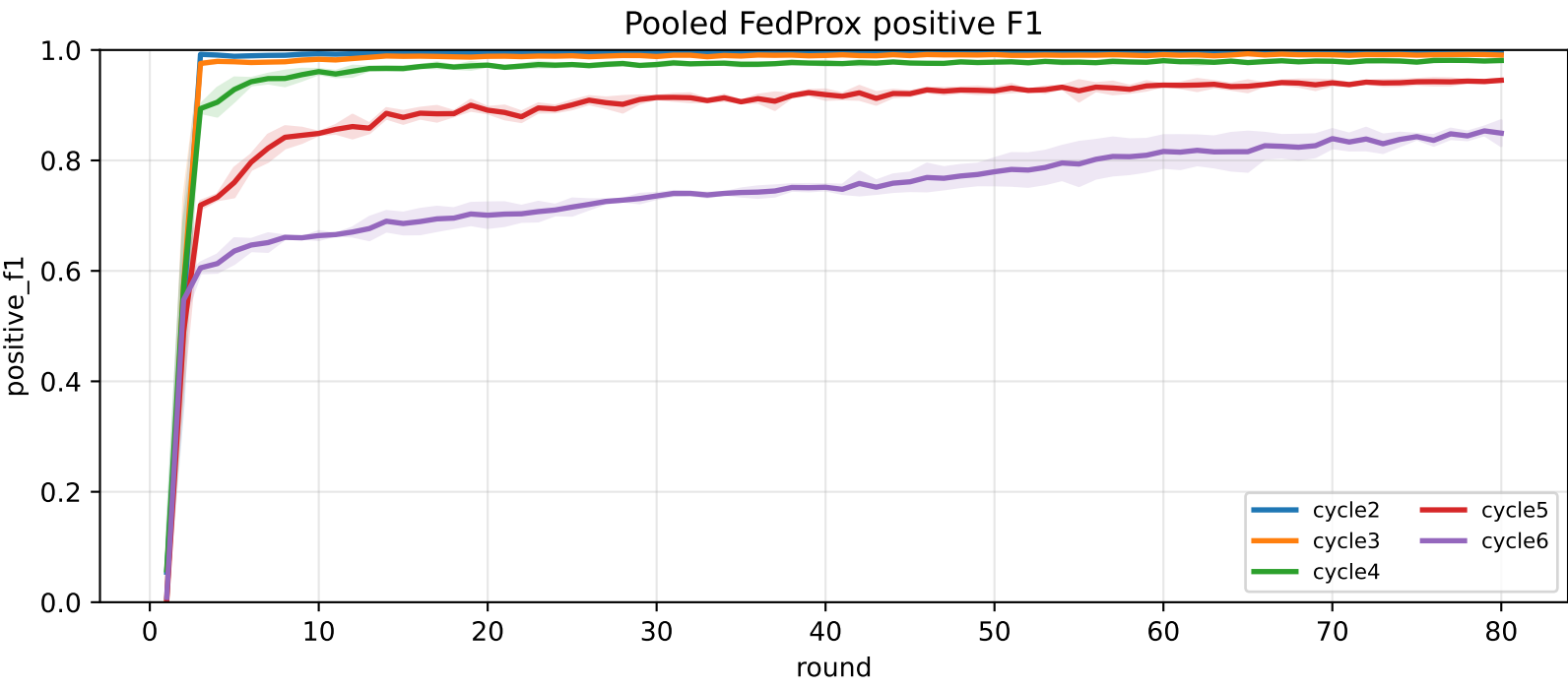
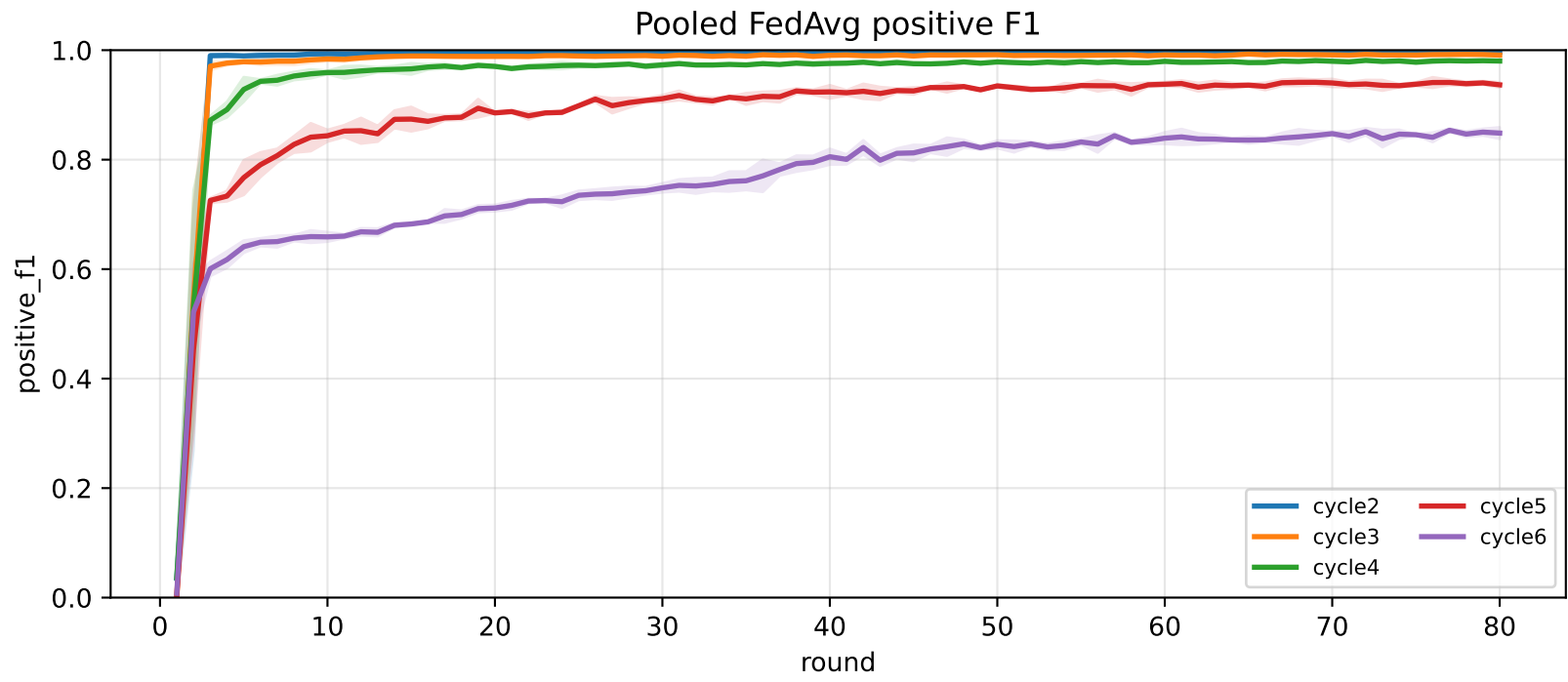
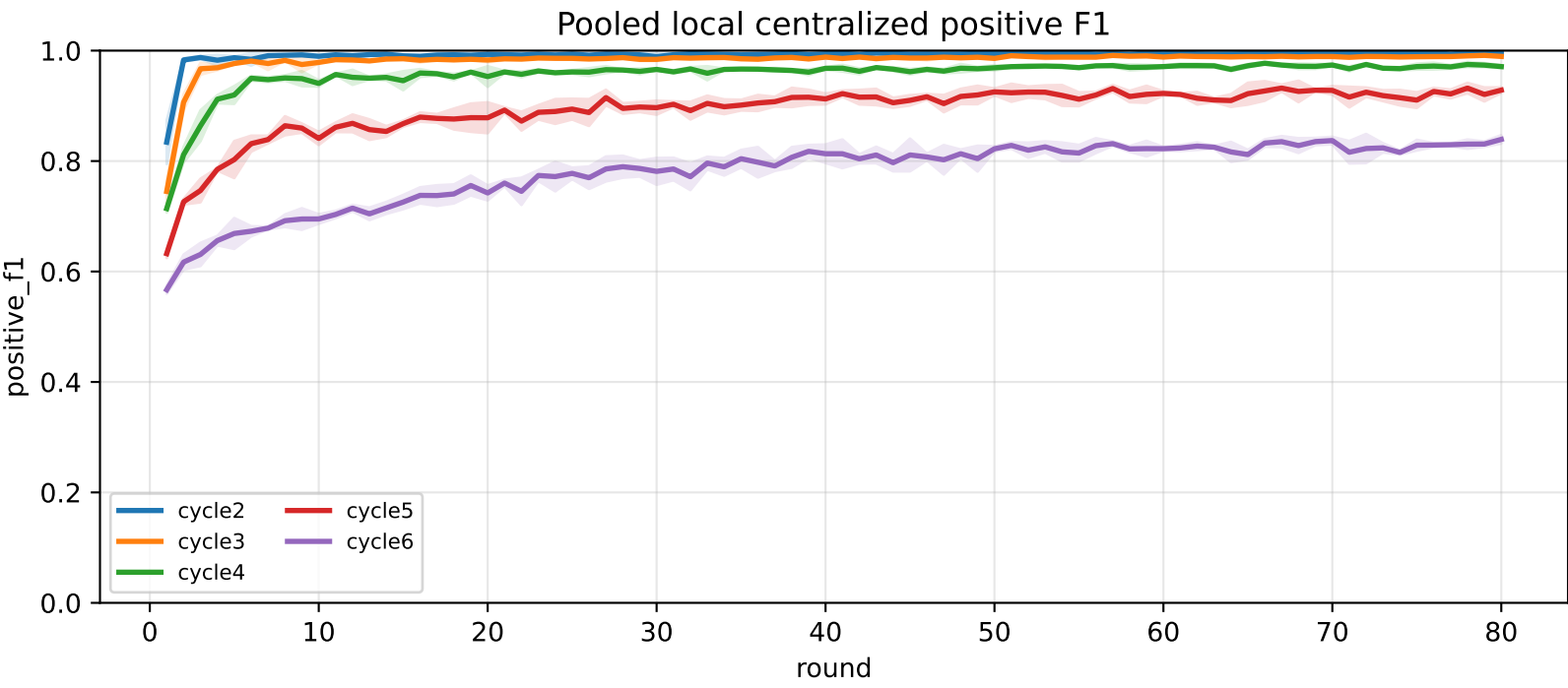
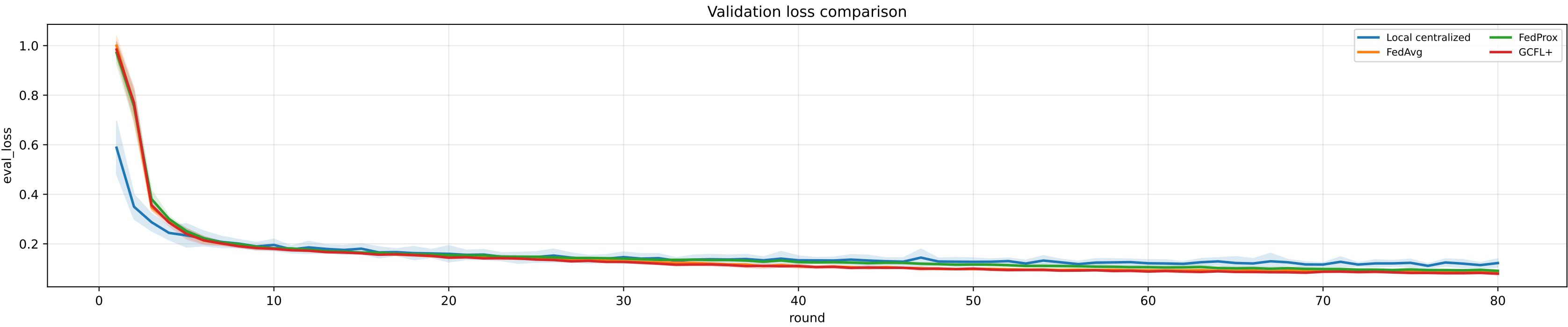
Best logged test performance (mean ± std over seeds)							
Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	92.1% ± 0.6%	85.7% ± 0.7%	99.6% ± 0.1%	98.9% ± 0.1%	95.0% ± 0.4%	79.3% ± 1.4%	55.6% ± 2.3%
FedAvg	94.1% ± 0.2%	88.5% ± 0.3%	99.8% ± 0.0%	99.4% ± 0.1%	96.9% ± 0.4%	85.2% ± 0.8%	61.5% ± 1.0%
FedProx	94.4% ± 0.6%	89.0% ± 0.8%	99.8% ± 0.0%	99.4% ± 0.0%	97.1% ± 0.3%	86.3% ± 0.6%	62.6% ± 3.6%
GCFL+	94.5% ± 0.3%	89.2% ± 0.4%	99.8% ± 0.0%	99.3% ± 0.1%	97.0% ± 0.2%	85.6% ± 0.3%	64.3% ± 1.5%



Heterogeneity level: High | family aggregated / pooled performance |
family=mild_decreasing | n_clients=4 |
model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64

family=mild_decreasing | cycle2=32.5%±0.7% | cycle3=30.7%±1.0% | cycle4=26.8%±1.1% | cycle5=24.5%±1.0% | cycle6=19.2%±1.0%
heterogeneity=high | task_jsd_mean=0.028 | task_jsd_median=0.018 | task_jsd_max=0.120 | family_centroid_jsd_mean=0.033
per-task pos-rate std | cycle2_std=0.120 | cycle3_std=0.071 | cycle4_std=0.032 | cycle5_std=0.054 | cycle6_std=0.087

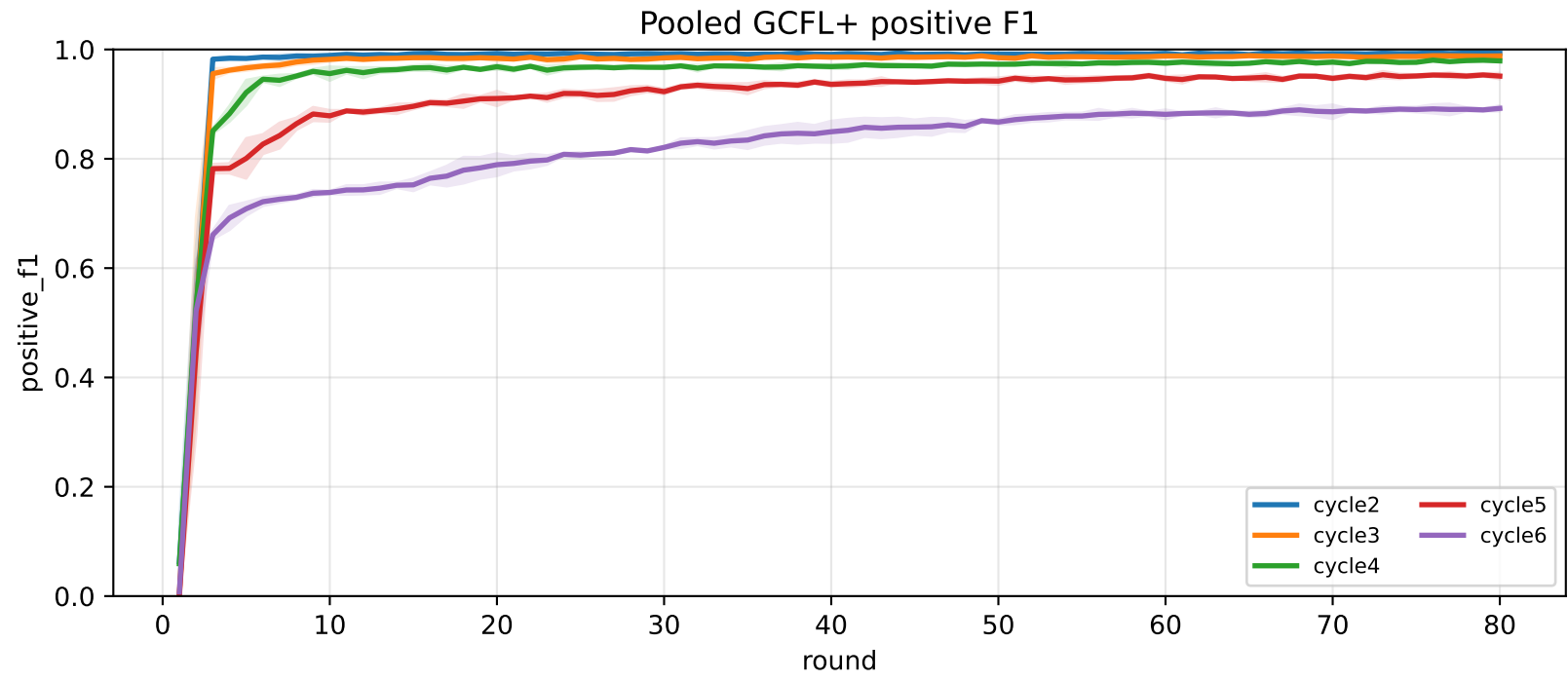
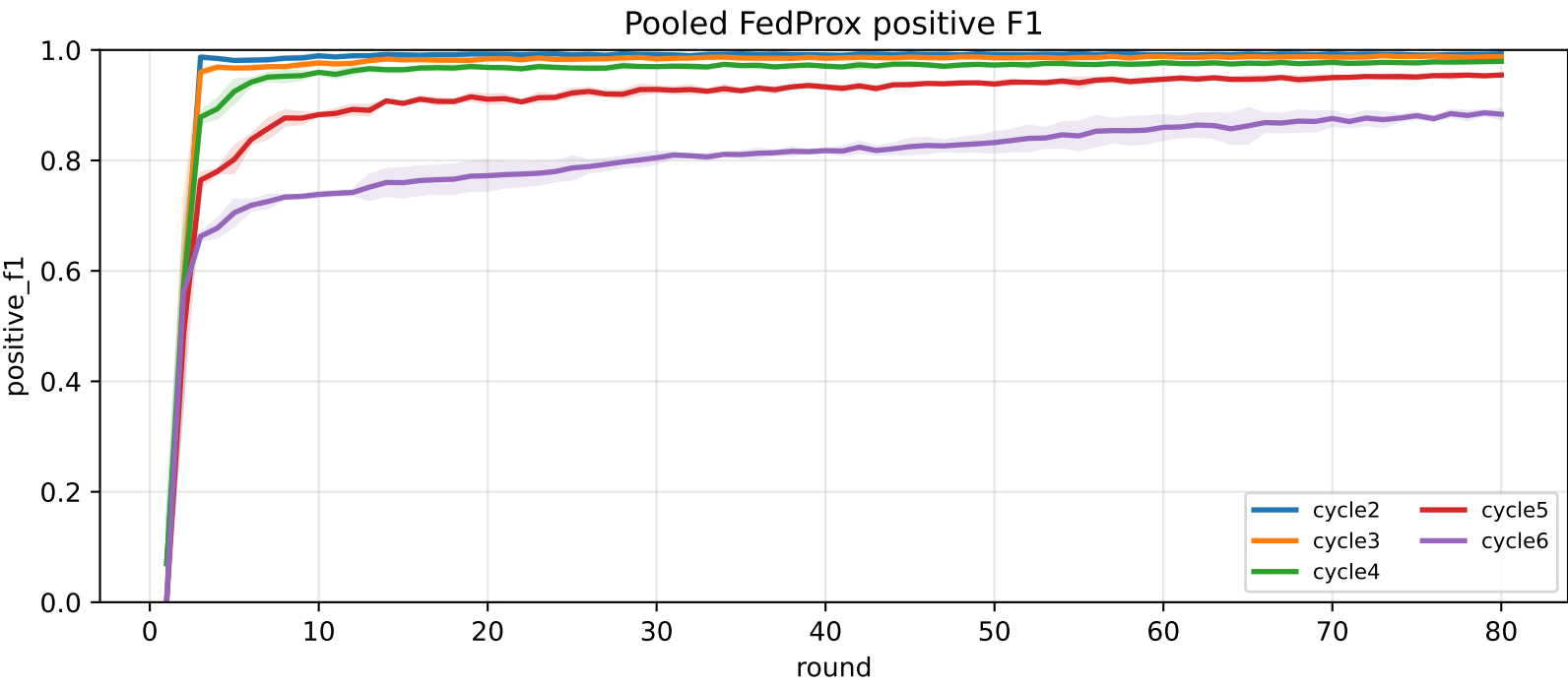
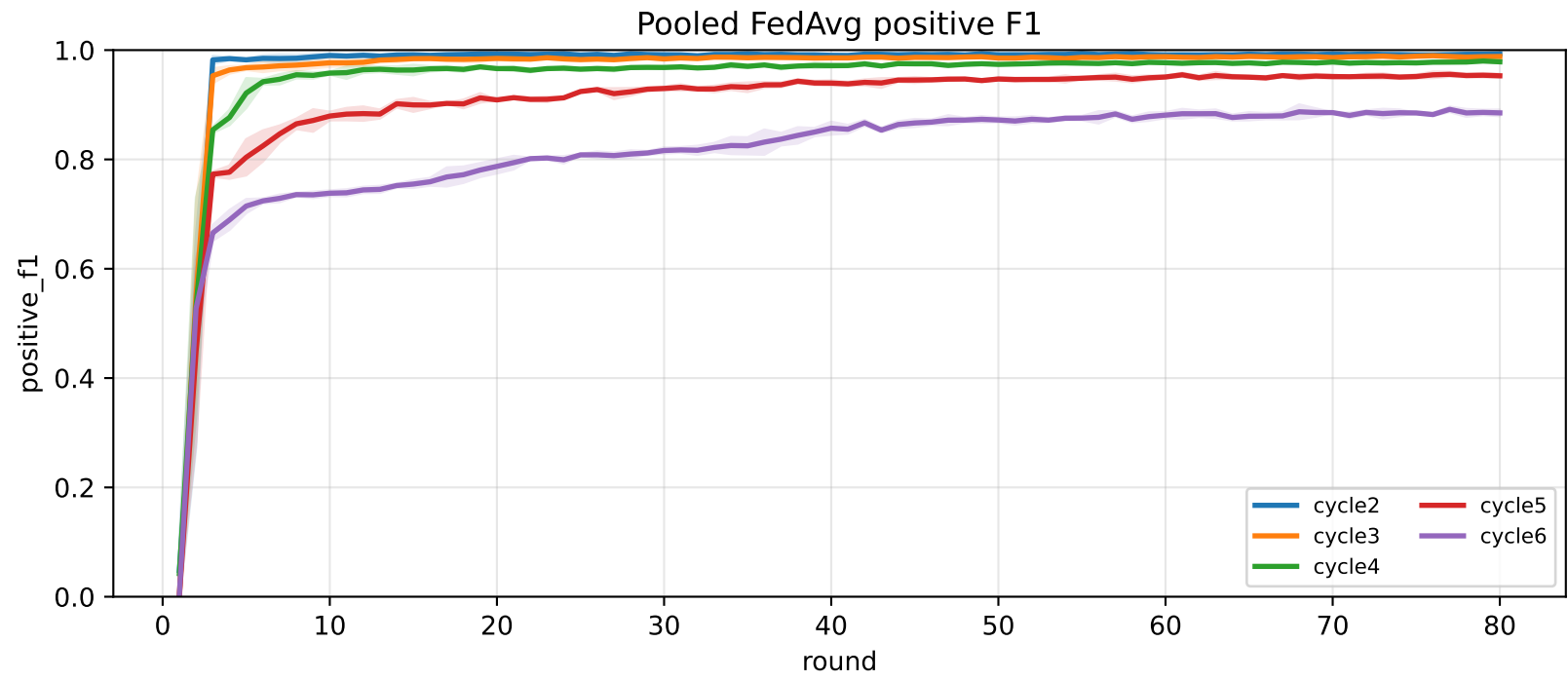
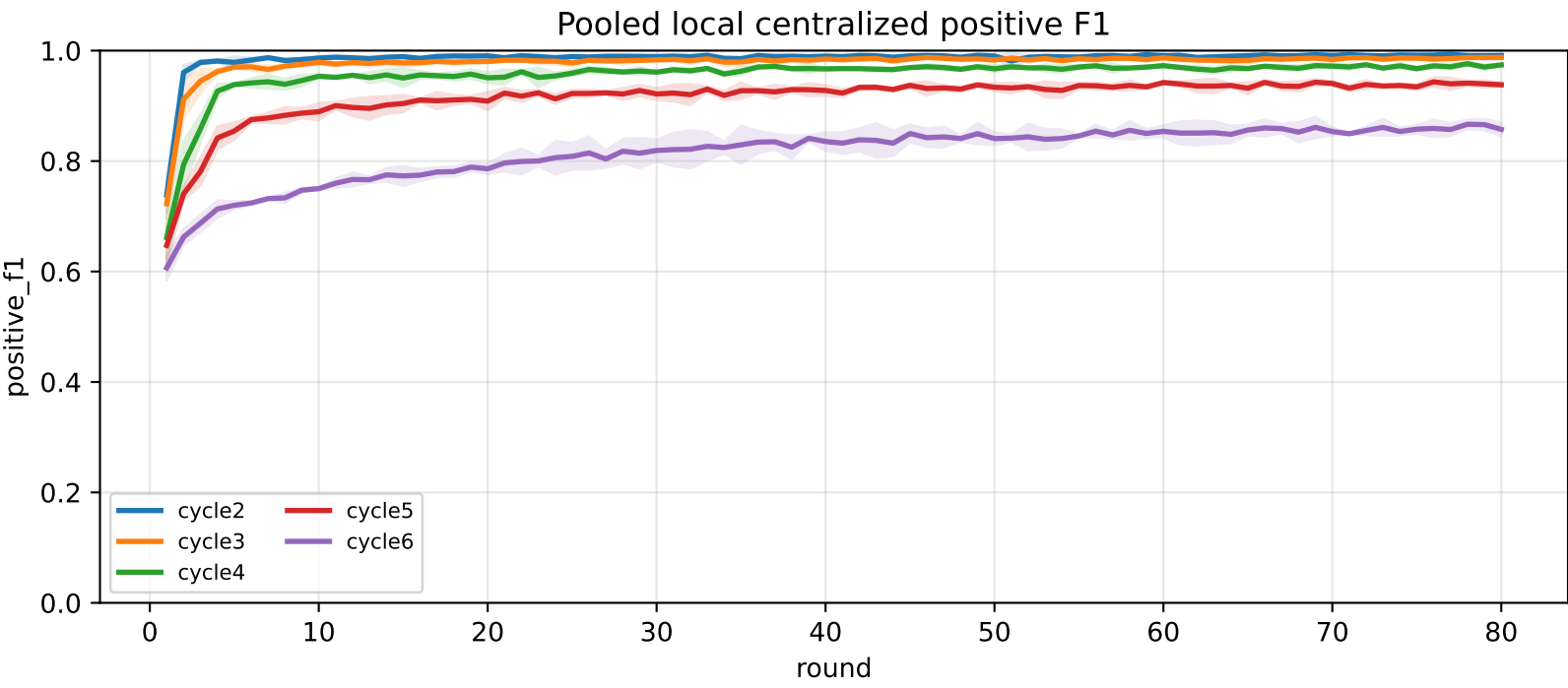
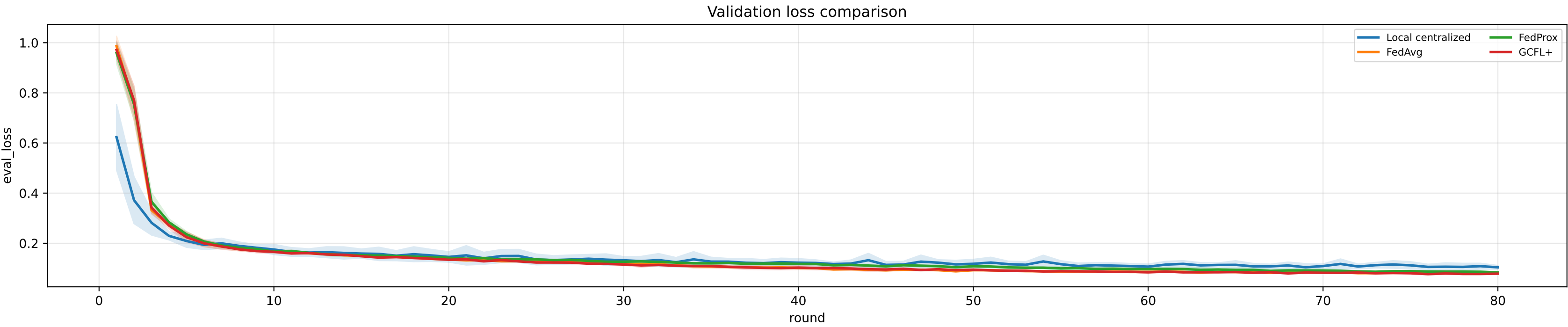
Best logged test performance (mean ± std over seeds)							
Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	95.2% ± 0.2%	94.4% ± 0.2%	99.5% ± 0.1%	99.1% ± 0.1%	97.3% ± 0.1%	92.9% ± 0.1%	83.2% ± 0.7%
FedAvg	95.9% ± 0.2%	95.2% ± 0.2%	99.6% ± 0.0%	99.3% ± 0.1%	98.0% ± 0.2%	94.5% ± 0.3%	84.7% ± 0.6%
FedProx	95.8% ± 0.4%	95.2% ± 0.4%	99.6% ± 0.0%	99.3% ± 0.1%	98.2% ± 0.2%	94.5% ± 0.2%	84.3% ± 1.6%
GCFL+	96.0% ± 0.1%	95.3% ± 0.2%	99.6% ± 0.1%	99.1% ± 0.1%	98.1% ± 0.1%	94.3% ± 0.1%	85.5% ± 0.7%



Heterogeneity level: High | family aggregated / pooled performance |
family=flat_balanced | n_clients=4 |
model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64

family=flat_balanced | cycle2=25.1%±0.7% | cycle3=26.6%±0.8% | cycle4=27.5%±0.8% | cycle5=27.8%±1.2% | cycle6=25.6%±0.3%
heterogeneity=high | task_jsd_mean=0.028 | task_jsd_median=0.018 | task_jsd_max=0.120 | family_centroid_jsd_mean=0.033
per-task pos-rate std | cycle2_std=0.120 | cycle3_std=0.071 | cycle4_std=0.032 | cycle5_std=0.054 | cycle6_std=0.087

Best logged test performance (mean ± std over seeds)							
Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	95.0% ± 0.1%	95.2% ± 0.1%	99.1% ± 0.1%	98.8% ± 0.2%	97.1% ± 0.1%	94.2% ± 0.2%	86.6% ± 0.5%
FedAvg	96.1% ± 0.1%	96.2% ± 0.1%	99.3% ± 0.0%	98.7% ± 0.0%	97.4% ± 0.1%	95.7% ± 0.2%	89.7% ± 0.5%
FedProx	95.9% ± 0.3%	96.0% ± 0.2%	99.3% ± 0.0%	98.7% ± 0.0%	97.5% ± 0.1%	95.6% ± 0.1%	89.1% ± 1.1%
GCFL+	96.0% ± 0.1%	96.2% ± 0.1%	99.3% ± 0.2%	98.7% ± 0.1%	97.4% ± 0.1%	95.3% ± 0.2%	90.1% ± 0.5%



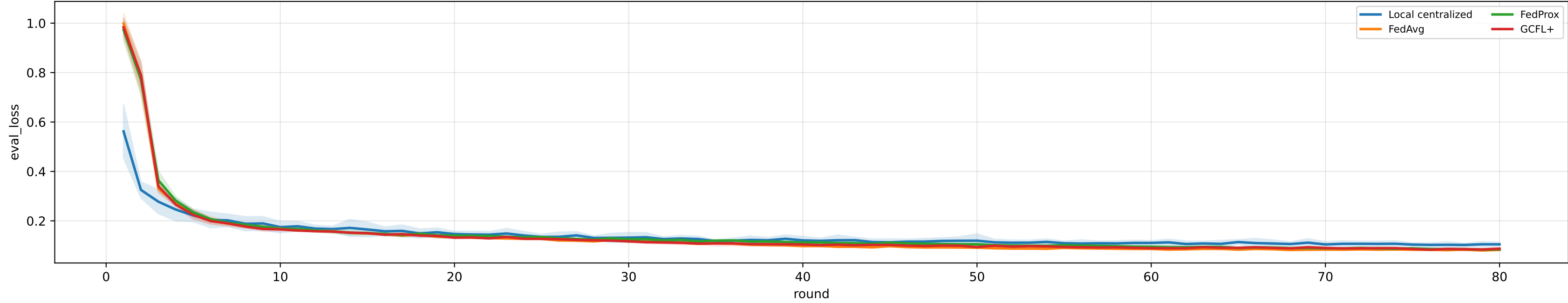
Heterogeneity level: High | family aggregated / pooled performance |
family=mild_increasing | n_clients=4 |
model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64

family=mild_increasing | cycle2=16.7%±0.8% | cycle3=21.4%±0.7% | cycle4=25.6%±1.2% | cycle5=28.8%±0.7% | cycle6=30.0%±0.4%
heterogeneity=high | task_jsd_mean=0.028 | task_jsd_median=0.018 | task_jsd_max=0.120 | family_centroid_jsd_mean=0.033
per-task pos-rate std | cycle2_std=0.120 | cycle3_std=0.071 | cycle4_std=0.032 | cycle5_std=0.054 | cycle6_std=0.087

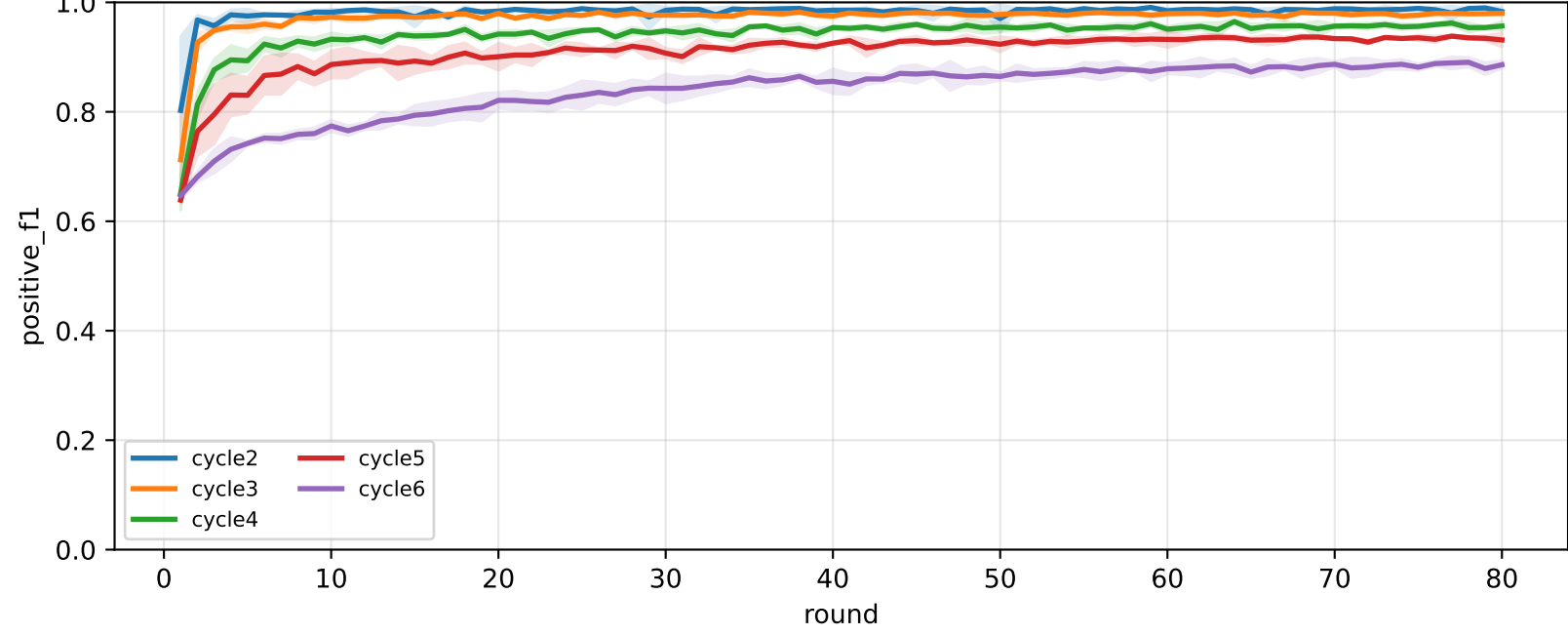
Best logged test performance (mean ± std over seeds)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	94.4% ± 0.1%	95.1% ± 0.1%	98.7% ± 0.2%	97.8% ± 0.2%	96.2% ± 0.2%	93.8% ± 0.4%	89.2% ± 0.5%
FedAvg	95.7% ± 0.1%	96.2% ± 0.1%	98.8% ± 0.2%	98.2% ± 0.2%	96.8% ± 0.1%	95.2% ± 0.2%	92.1% ± 0.0%
FedProx	95.6% ± 0.2%	96.1% ± 0.2%	98.8% ± 0.2%	98.1% ± 0.1%	96.8% ± 0.2%	95.0% ± 0.3%	91.6% ± 0.5%
GCFL+	95.4% ± 0.2%	95.9% ± 0.2%	98.7% ± 0.1%	98.1% ± 0.2%	96.5% ± 0.2%	94.3% ± 0.4%	91.9% ± 0.3%

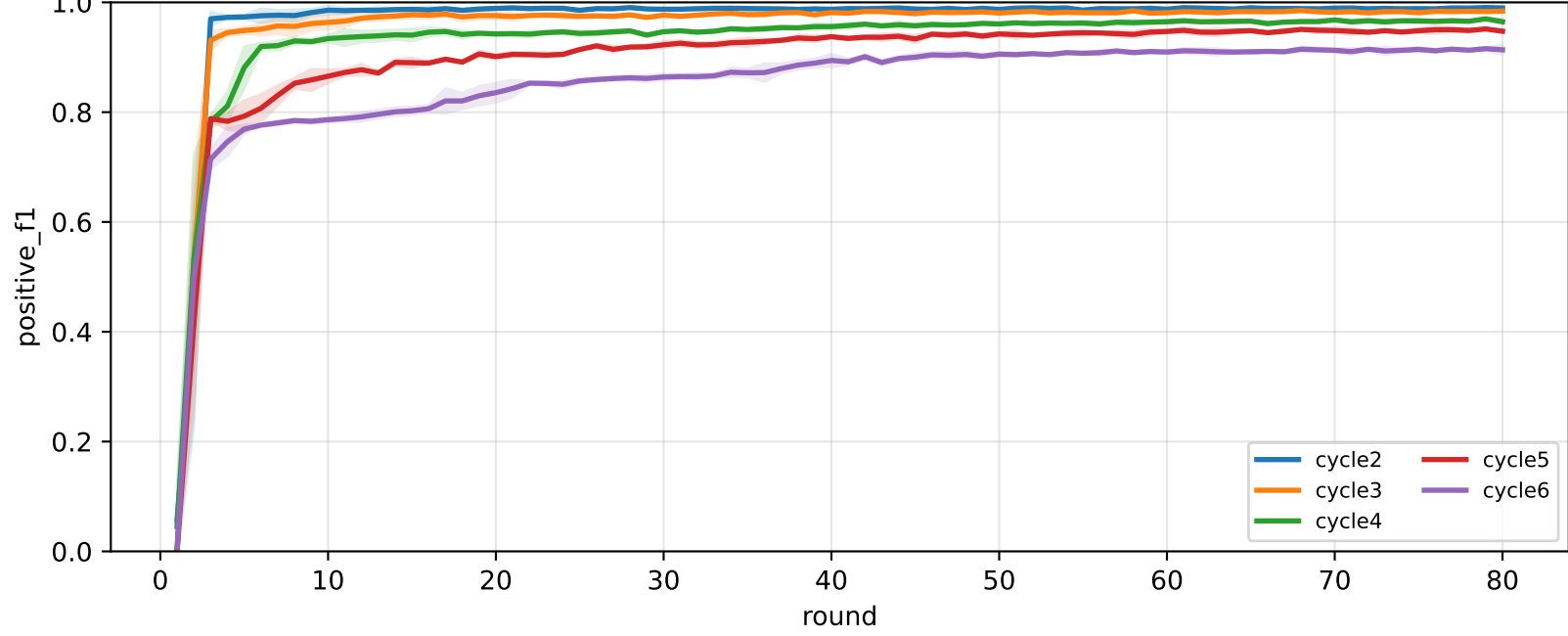
Validation loss comparison



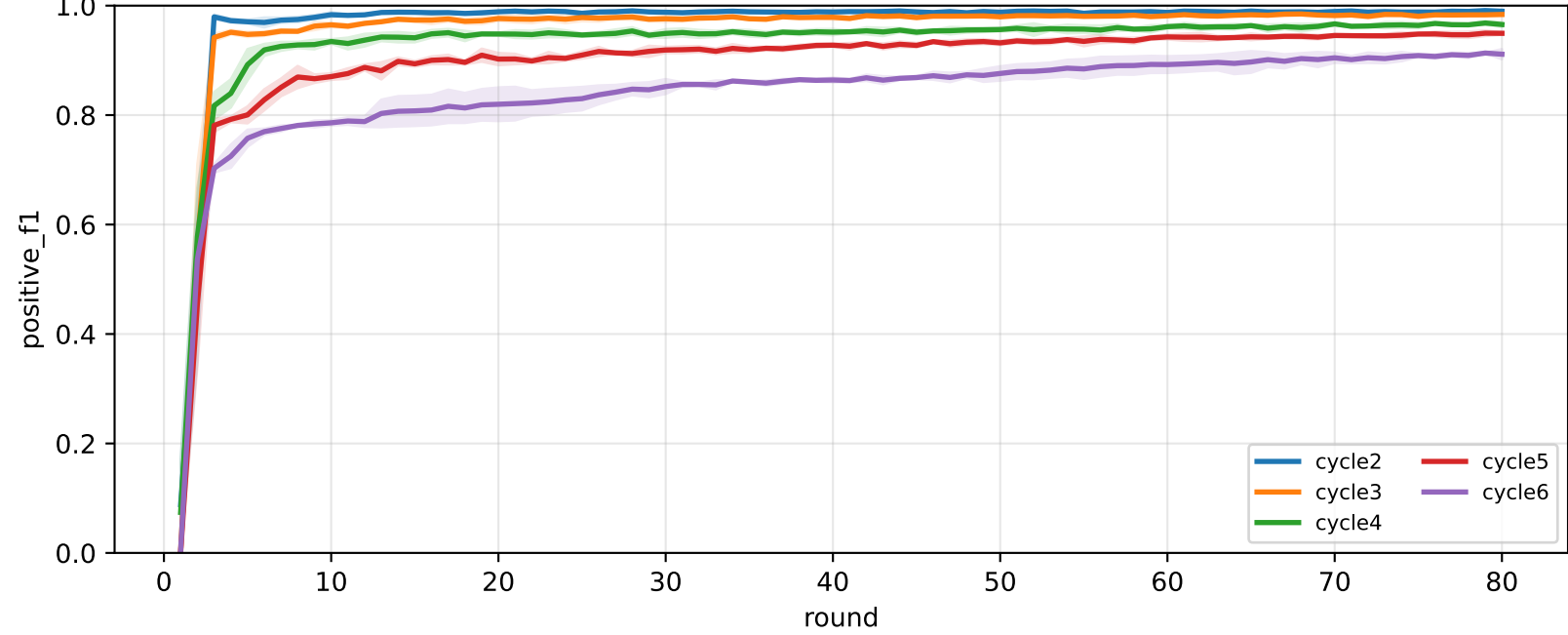
Pooled local centralized positive F1



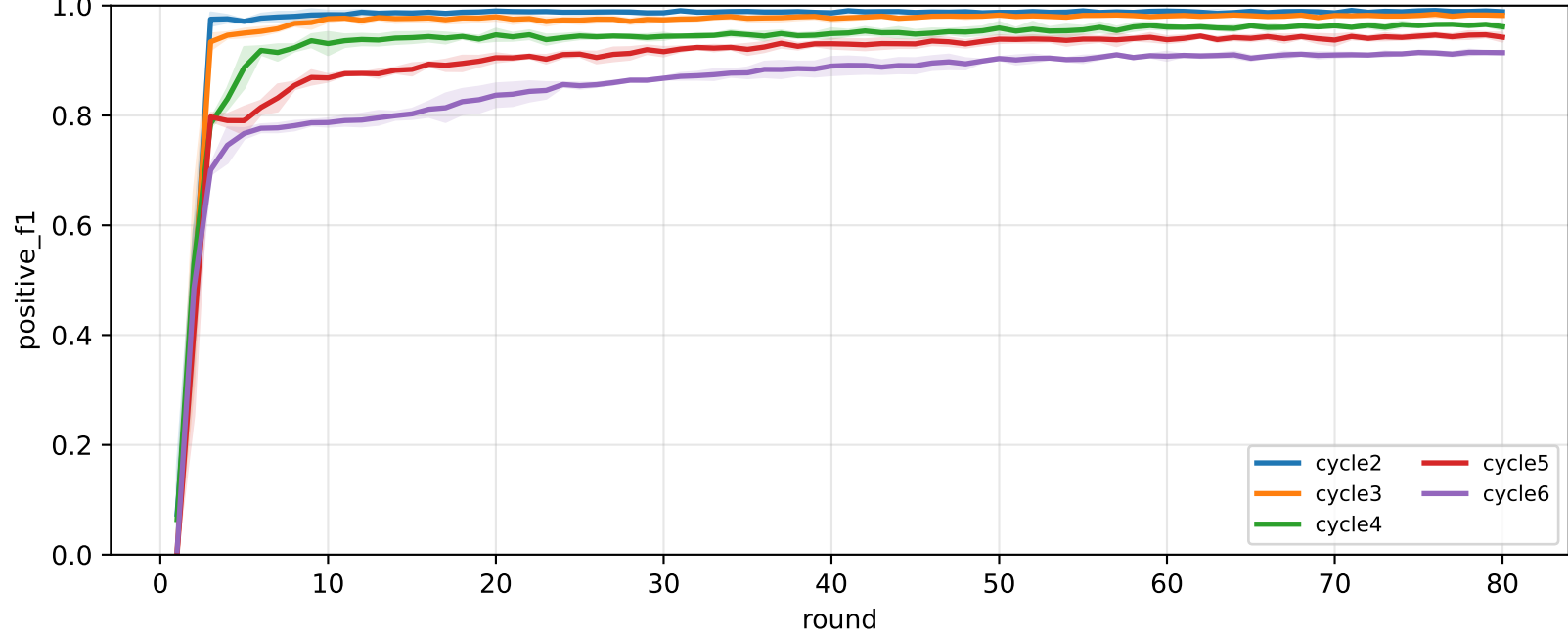
Pooled FedAvg positive F1



Pooled FedProx positive F1



Pooled GCFL+ positive F1



Heterogeneity level: High | family aggregated / pooled performance |
family=strong_increasing | n_clients=4 |
model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64

family=strong_increasing | cycle2=9.9%±1.2% | cycle3=14.1%±0.8% | cycle4=19.4%±1.7% | cycle5=25.4%±1.6% | cycle6=31.3%±1.0%
heterogeneity=high | task_jsd_mean=0.028 | task_jsd_median=0.018 | task_jsd_max=0.120 | family_centroid_jsd_mean=0.033
per-task pos-rate std | cycle2_std=0.120 | cycle3_std=0.071 | cycle4_std=0.032 | cycle5_std=0.054 | cycle6_std=0.087

Best logged test performance (mean ± std over seeds)							
Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	92.3% ± 0.2%	93.6% ± 0.2%	98.2% ± 0.1%	96.3% ± 0.1%	94.2% ± 0.0%	91.2% ± 0.2%	88.2% ± 0.7%
FedAvg	93.1% ± 0.1%	94.2% ± 0.1%	98.2% ± 0.1%	96.5% ± 0.1%	94.9% ± 0.1%	92.7% ± 0.4%	89.0% ± 0.1%
FedProx	92.8% ± 0.2%	94.0% ± 0.2%	98.2% ± 0.1%	96.4% ± 0.1%	94.6% ± 0.3%	92.4% ± 0.3%	88.6% ± 0.4%
GCFL+	91.7% ± 0.4%	93.1% ± 0.3%	97.9% ± 0.1%	95.8% ± 0.3%	93.9% ± 0.4%	91.1% ± 0.5%	86.8% ± 0.5%

